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JOULE-THOMSON CRYOCOOLER

Abstract

A Joule-Thomson cryocooler includes a pre-cooling cryocooler that includes a pre-cooling stage, a stage extension component that is attached to the pre-cooling stage and is cooled by the pre-cooling stage, and a refrigerant pipe that is mounted on the stage extension component to enable heat exchange with the stage extension component.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This is a bypass continuation of International PCT Application No. PCT/JP2023/038530, filed on Oct. 25, 2023, which claims priority to Japanese Patent Application No. 2022-188372, filed on Nov. 25, 2022, which are incorporated by reference herein in their entirety.

BACKGROUND

Technical Field

[0002] A certain embodiment of the present invention relates to a Joule-Thomson (JT) cryocooler.

Description of Related Art

[0003] In the related art, there has been known a JT cryocooler that includes a JT valve that enables cooling of a refrigerant gas using JT expansion, and a pre-cooling cryocooler such as a Gifford-McMahon (GM) cryocooler that pre-cools the refrigerant gas supplied to the JT valve.

SUMMARY

[0004] According to an embodiment of the present invention, there is provided a Joule-Thomson cryocooler including a pre-cooling cryocooler that includes a pre-cooling stage, a stage extension component that is attached to the pre-cooling stage and is cooled by the pre-cooling stage, and a refrigerant pipe that is mounted on the stage extension component to enable heat exchange with the stage extension component.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] FIG. 1 is a diagram schematically showing a cryogenic cooling device according to an embodiment.

[0006] FIG. 2 is a diagram schematically showing another example of a stage extension component according to the embodiment.

DETAILED DESCRIPTION

[0007] In general, a refrigerant pipe of a JT cryocooler is mounted by being directly wound around a cooling stage of a pre-cooling cryocooler. A refrigerant gas flowing through the refrigerant pipe is cooled to a target pre-cooling temperature by heat exchange with the cooling stage. However, in a case where the cooling stage is small in size and has a small surface area, a heat exchange area on which the refrigerant pipe is mounted may be insufficient, and the pre-cooling of the refrigerant gas may be insufficient.

[0008] It is desirable to expand the heat exchange area of the pre-cooling cryocooler of the JT cryocooler.

[0009] Hereinafter, an embodiment for carrying out the present invention will be described in detail with reference to the drawings. In the description and the drawings, the same or equivalent components, members, and processes are denoted by the same reference numerals, and overlapping description is omitted as appropriate. The scale and the shape of each of parts illustrated in the drawings are set for convenience to make the description easy to understand, and are not to be interpreted as limiting unless stated otherwise. The embodiment is merely an example and does not limit the scope of the present invention. All features described in the embodiment or combinations thereof are not necessarily essential to the present invention.

[0010] FIG. 1 is a diagram schematically showing a cryogenic cooling device **10** according to an embodiment. The cryogenic cooling device **10** includes a vacuum chamber **12**, a radiation shield **14**, and a JT cryocooler **18** for cooling an object **16** to be cooled.

[0011] The vacuum chamber **12** may be, for example, a cryostat, and is configured to provide a

cryogenic vacuum environment therein. The vacuum chamber **12** is formed of a metal material such as stainless steel or other suitable high-strength material to withstand ambient pressure (for example, atmospheric pressure). The radiation shield **14**, a low-temperature section of the JT cryocooler **18**, and the object **16** to be cooled are disposed in the vacuum chamber **12**.

[0012] The radiation shield **14** is disposed in the vacuum chamber **12** to surround the low-temperature section of the JT cryocooler **18** and the object **16** to be cooled, and suppresses inflow of radiant heat from the vacuum chamber **12** to the JT cryocooler **18** and the object **16** to be cooled. The radiation shield **14** is formed of a high thermal conductivity metal material such as copper (for example, pure copper). An insulator such as a multi-layer insulator may be disposed between the vacuum chamber **12** and the radiation shield **14**.

[0013] The object to be cooled **16** may be, for example, a superconducting device such as a superconducting coil, a measuring device that operates better at a cryogenic temperature, or other devices for use at a cryogenic temperature. Alternatively, the object **16** to be cooled may be, for example, a cryogenic fluid such as liquid helium, and the JT cryocooler **18** may be used for recondensing a vaporized cryogenic fluid.

[0014] The JT cryocooler **18** includes a pre-cooling cryocooler **20** and a refrigerant circuit **40** including a JT valve **30** and a final heat exchanger **32**. A refrigerant flowing through the refrigerant circuit **40** is pre-cooled by the pre-cooling cryocooler **20**, further cooled by JT expansion in the JT valve **30**, and is supplied to the final heat exchanger **32**. The object **16** to be cooled is cooled by heat exchange with the final heat exchanger **32**. The cooled refrigerant is collected from the final heat exchanger **32**, pressurized by a compressor described below, pre-cooled by the pre-cooling cryocooler **20** again, and is supplied to the JT valve **30**. In this way, the refrigerant circulates through the refrigerant circuit **40**. The JT cryocooler **18** is capable of cooling the final heat exchanger **32** to a temperature range of, for example, about 4 K or lower (for example, 1 K to 4 K), and thus, the object **16** to be cooled can be cooled to the temperature range.

[0015] The pre-cooling cryocooler **20** is, for example, a two-stage GM cryocooler. The pre-cooling cryocooler **20** includes a first compressor **21** and an expander **22** also referred to as a cold head. The expander **22** includes a drive unit **23**, a first cylinder **24**, a first pre-cooling stage **25**, a second cylinder **26**, and a second pre-cooling stage **27**. The first compressor **21** is disposed in an ambient environment (for example, room temperature and atmospheric pressure environment), that is, outside the vacuum chamber **12**. The expander **22** is installed in the vacuum chamber **12** such that the drive unit **23** is disposed outside the vacuum chamber **12** and the cylinders and the pre-cooling stages are disposed inside the vacuum chamber **12**.

[0016] The first cylinder **24** connects the first pre-cooling stage **25** to the drive unit **23**, so that the first pre-cooling stage **25** is structurally supported by the drive unit **23**. The second cylinder **26** connects the second pre-cooling stage **27** to the first pre-cooling stage **25**, so that the second pre-cooling stage **27** is structurally supported by the first pre-cooling stage **25**. The first cylinder **24** and the second cylinder **26** extend coaxially, and the drive unit **23**, the first cylinder **24**, the first pre-cooling stage **25**, the second cylinder **26**, and the second pre-cooling stage **27** are linearly arranged in a line in this order. Typically, the first pre-cooling stage **25** and the second pre-cooling stage **27** are formed of a high thermal conductivity metal material such as copper (for example, pure copper), and the first cylinder **24** and the second cylinder **26** are formed of other metal materials such as stainless steel.

[0017] A first displacer and a second displacer (not shown) are disposed reciprocally in an inside of each of the first cylinder **24** and the second cylinder **26**. A first regenerator and a second regenerator (not shown) are incorporated into the first displacer and the second displacer, respectively. In addition, the drive unit **23** includes a drive mechanism (not shown) such as a motor for reciprocating the first displacer and the second displacer. The drive mechanism includes a flow path switching mechanism that switches a flow path of a refrigerant gas to periodically repeat supply and exhaust of the refrigerant gas to and from an inside of the expander **22**. A refrigerant gas

of the pre-cooling cryocooler **20** is usually a helium gas, but other appropriate gases may be used. [0018] The first compressor **21** is configured to collect the refrigerant gas from the expander **22**, to pressurize the collected refrigerant gas, and to supply the refrigerant gas to the expander **22** again. By circulating the refrigerant gas between the first compressor **21** and the expander **22** with an appropriate combination of a pressure fluctuation and a volume fluctuation of the refrigerant gas in the expander **22**, a thermodynamic cycle (for example, a GM cycle) for cold generation is configured, and the expander **22** can provide cryogenic cooling.

[0019] The first pre-cooling stage **25** is cooled to a first cooling temperature, and the second pre-cooling stage **27** is cooled to a second cooling temperature lower than the first cooling temperature. The first cooling temperature may be selected from a temperature range of, for example, 50 K or higher and 150 K or lower. The second cooling temperature may be selected from a temperature range of, for example, 10 K or higher and 25 K or lower.

[0020] In this embodiment, a stage extension component **50** is attached to the first pre-cooling stage **25**. The stage extension component **50** is thermally coupled to the first pre-cooling stage **25** and is cooled to the first cooling temperature by the first pre-cooling stage **25**. Therefore, the stage extension component **50** can be integrated with the first pre-cooling stage **25** and can function as a part of the first pre-cooling stage **25**.

[0021] The stage extension component **50** is formed of, for example, a metal material having a high thermal conductivity, such as copper (for example, pure copper or a copper alloy) or aluminum (for example, pure aluminum or an aluminum alloy), or other high thermal conductivity material. Here, the high thermal conductivity material may be, for example, a material having a higher thermal conductivity than stainless steel (for example, SUS304).

[0022] The stage extension component **50** has a tubular (for example, cylindrical) shape, is fixed to the first pre-cooling stage **25** at one end, and extends toward the second pre-cooling stage **27**. The second cylinder **26** is disposed in the stage extension component **50**. A length in an axial direction of the stage extension component **50** is shorter than that of the second cylinder **26**. Therefore, the other end of the stage extension component **50** does not reach the second pre-cooling stage **27**. Therefore, the stage extension component **50** is not in physical contact with the second pre-cooling stage **27**.

[0023] The stage extension component **50** is detachably attached to the first pre-cooling stage **25**. For example, a flange formed at one end of the stage extension component **50** may be fixed to the first pre-cooling stage **25** by a fastening member such as a bolt. By detaching the fastening member, the stage extension component **50** can be released from the first pre-cooling stage **25**. The stage extension component **50** may be attached to the first pre-cooling stage **25** by a non-detachable method such as brazing or welding.

[0024] As shown in FIG. **1**, the stage extension component **50** may connect the first pre-cooling stage **25** to the radiation shield **14**. For example, a flange formed at an end portion of the stage extension component **50** on a side opposite to the first pre-cooling stage **25** may be fixed to the radiation shield **14** by a fastening member such as a bolt. The radiation shield **14** may be thermally coupled to the first pre-cooling stage **25** via the stage extension component **50** and may be cooled to the first cooling temperature by the first pre-cooling stage **25**. The radiation shield **14** surrounds the low-temperature section of the JT cryocooler **18**, such as the second pre-cooling stage **27**, and can suppress heat inflow to the low-temperature section.

[0025] The radiation shield **14** may be directly attached to the first pre-cooling stage **25** without using the stage extension component **50**, or may be thermally coupled to the first pre-cooling stage **25** via a heat transfer member separate from the stage extension component **50**.

[0026] The refrigerant circuit **40** includes a second compressor **41**, a heat exchanger group **42**, and a refrigerant supply line **44** and a refrigerant collection line **46** that connect these components in addition to the JT valve **30** and the final heat exchanger **32**. A refrigerant gas that circulates through the refrigerant circuit **40** is usually a helium gas, but other appropriate gases may be used. The

refrigerant circuit **40** is not limited to the specific configuration described herein and can adopt various typical configurations as appropriate.

[0027] The second compressor **41** is configured to pressurize the refrigerant gas collected from the refrigerant collection line **46** and to send the collected refrigerant gas to the refrigerant supply line **44**. For understanding, a direction in which the refrigerant flows is indicated by an arrow in FIG. **1**. The second compressor **41** serves as a refrigerant source that circulates the refrigerant in the refrigerant circuit **40**. The second compressor **41** is disposed outside the vacuum chamber **12**.

[0028] The heat exchanger group **42** in the refrigerant circuit **40** is disposed between the second compressor **41** and the final heat exchanger **32**. The heat exchanger group **42** is composed of a series of counterflow heat exchangers (**42a** to **42c**), and has a three-stage configuration of a first heat exchanger **42a**, a second heat exchanger **42b**, and a third heat exchanger **42c** in the present embodiment. The first heat exchanger **42a** is disposed between the vacuum chamber **12** and the radiation shield **14**, that is, in a space inside the vacuum chamber **12** and outside the radiation shield **14**. The second heat exchanger **42b**, the third heat exchanger **42c**, and the final heat exchanger **32** are disposed inside the radiation shield **14**.

[0029] The first heat exchanger **42a** cools a high-temperature (for example, a room temperature, for example, approximately 300 K) refrigerant gas flowing into the vacuum chamber **12** from the outside of the vacuum chamber **12**. The second heat exchanger **42b** further cools the refrigerant cooled by the first heat exchanger **42a** and the first pre-cooling stage **25**. The third heat exchanger **42c** further cools the refrigerant cooled by the second heat exchanger **42b** and the second pre-cooling stage **27**.

[0030] The refrigerant supply line **44** connects a discharge side of the second compressor **41** to a refrigerant inlet of the final heat exchanger **32**, and the refrigerant collection line **46** connects a refrigerant outlet of the final heat exchanger **32** to a suction side of the second compressor **41**. The refrigerant supply line **44** includes a high pressure side flow path of each of the first heat exchanger **42a**, the second heat exchanger **42b**, and the third heat exchanger **42c**, and the refrigerant collection line **46** includes a low pressure side flow path of each of the first heat exchanger **42a**, the second heat exchanger **42b**, and the third heat exchanger **42c**. The refrigerant flowing through the high pressure side flow path can be cooled by heat exchange between the high pressure side flow path and the low pressure side flow path in each heat exchanger. The high pressure side flow path and the low pressure side flow path may be called a high temperature side flow path and a low temperature side flow path, respectively.

[0031] In addition, the refrigerant supply line **44** includes a first refrigerant pipe **44a** and a second refrigerant pipe **44b**. The refrigerant pipes are formed of, for example, a high thermal conductivity metal material such as copper (for example, pure copper).

[0032] The first refrigerant pipe **44a** extends from the first heat exchanger **42a** via the first pre-cooling stage **25** to the second heat exchanger **42b**. The first refrigerant pipe **44a** connects the high pressure side flow path of the first heat exchanger **42a** to the high pressure side flow path of the second heat exchanger **42b**. The first refrigerant pipe **44a** is thermally coupled to the first pre-cooling stage **25**, and the refrigerant flowing through the first refrigerant pipe **44a** is cooled by the first pre-cooling stage **25**.

[0033] Note that, in this embodiment, the first refrigerant pipe **44a** is mounted on the stage extension component **50** to enable heat exchange with the stage extension component **50**, instead of the first pre-cooling stage **25**. For example, the first refrigerant pipe **44a** is fixed to the stage extension component **50** in a state of being wound around an outer peripheral surface of the stage extension component **50**. The first refrigerant pipe **44a** is not wound around the first pre-cooling stage **25**. As described above, the stage extension component **50** is attached and thermally coupled to the first pre-cooling stage **25**, so that the first refrigerant pipe **44a** is cooled to the first cooling temperature by the stage extension component **50** cooled by the first pre-cooling stage **25**.

[0034] The second refrigerant pipe **44b** extends from the second heat exchanger **42b** via the second

pre-cooling stage 27 to the third heat exchanger 42c. The second refrigerant pipe 44b connects the high pressure side flow path of the second heat exchanger 42b to the high pressure side flow path of the third heat exchanger 42c. The second refrigerant pipe 44b is thermally coupled to the second pre-cooling stage 27, and the refrigerant flowing through the second refrigerant pipe 44b is cooled by the second pre-cooling stage 27. The second refrigerant pipe 44b may be fixed to the second pre-cooling stage 27 in a state of being wound around an outer peripheral surface of the second pre-cooling stage 27.

[0035] The JT valve 30 is disposed between the last heat exchanger of the heat exchanger group 42 (in the present example, the third heat exchanger 42c) and the final heat exchanger 32 in the refrigerant supply line 44. The high pressure side flow path of the third heat exchanger 42c is connected to the refrigerant inlet of the final heat exchanger 32 via the JT valve 30. The JT valve 30 is a fixed orifice in the present embodiment. However, the JT valve 30 may be a variable orifice of which an opening degree is adjustable.

[0036] In the steady operation of the JT cryocooler 18, the refrigerant flows through the refrigerant circuit 40 as follows. The high-pressure refrigerant compressed by the second compressor 41 is first supplied to the high pressure side flow path of the first heat exchanger 42a. The high-pressure refrigerant flowing through the high pressure side flow path of the first heat exchanger 42a is cooled by heat exchange with the returning low-pressure refrigerant flowing through the low pressure side flow path of the first heat exchanger 42a. The high-pressure refrigerant cooled by the first heat exchanger 42a flows into the first refrigerant pipe 44a.

[0037] The high-pressure refrigerant flowing through the first refrigerant pipe 44a is cooled by the first pre-cooling stage 25 of the pre-cooling cryocooler 20 and is sent into the high pressure side flow path of the second heat exchanger 42b. The high-pressure refrigerant flowing through the high pressure side flow path of the second heat exchanger 42b is cooled by heat exchange with the returning low-pressure refrigerant flowing through the low pressure side flow path of the second heat exchanger 42b. The high-pressure refrigerant cooled by the second heat exchanger 42b flows into the second refrigerant pipe 44b.

[0038] The high-pressure refrigerant flowing through the second refrigerant pipe 44b is cooled by the second pre-cooling stage 27 of the pre-cooling cryocooler 20 and is sent into the high pressure side flow path of the third heat exchanger 42c. The high-pressure refrigerant flowing through the high pressure side flow path of the third heat exchanger 42c is cooled by heat exchange with the returning low-pressure refrigerant flowing through the low pressure side flow path of the third heat exchanger 42c. In this way, the high-pressure refrigerant is cooled to a temperature (that is, a temperature equal to or lower than an inversion temperature) at which the JT effect is expected, and is sent to the JT valve 30.

[0039] When the cooled high-pressure refrigerant passes through the JT valve 30, the cooled high-pressure refrigerant becomes a low-pressure refrigerant in a mist-like gas-liquid mixed state due to the Joule-Thomson effect, and generates a cooling capacity in a temperature range of the liquefied refrigerant. The mist-like low-pressure refrigerant is sent to the final heat exchanger 32. As described above, in a case where the refrigerant is helium, the final heat exchanger 32 can be cooled to a liquid helium temperature range. The final heat exchanger 32 can cool the object 16 to be cooled to the temperature through heat exchange with the object 16 to be cooled.

[0040] When cooling the final heat exchanger 32, the mist-like low-pressure refrigerant evaporates and vaporizes again. The unliquefied refrigerant and the refrigerant that is vaporized by evaporation in the JT valve 30 are returned to the low pressure side flow path of the third heat exchanger 42c. The low-pressure refrigerant flows through the refrigerant collection line 46 in the order of the third heat exchanger 42c, the second heat exchanger 42b, and the first heat exchanger 42a. In this case, the low-pressure refrigerant is heated while cooling the high-pressure refrigerant in each of the heat exchangers (42c, 42b, and 42a) as described above. In this way, the low-pressure refrigerant that has returned to the room temperature exits the vacuum chamber 12, is

collected in the second compressor **41**, and is compressed again.

[0041] In this way, the cryogenic cooling device **10** can cool the object **16** to be cooled to a desired temperature lower than the second cooling temperature of the pre-cooling cryocooler **20**, for example, about 4 K or lower (for example, 1 K to 4 K).

[0042] According to the embodiment, the stage extension component **50** is attached to the first pre-cooling stage **25**, thereby expanding a heat exchange area of one stage of the pre-cooling cryocooler **20**. By adjusting the shape of the stage extension component **50**, it is possible to secure an area required to sufficiently pre-cool the first refrigerant pipe **44a**. For example, a surface area of the stage extension component **50** can be increased by increasing dimensions of the stage extension component **50**, such as by increasing its length in the axial direction or increasing its diameter. Even when the first pre-cooling stage **25** is small in size and has a small surface area, the stage extension component **50** can compensate for the lack of the heat exchange area for the first refrigerant pipe **44a**.

[0043] As a result, the pre-cooling cryocooler **20** can adopt a general-purpose product. For example, a commercially available GM cryocooler can be used as the pre-cooling cryocooler **20**. It is no longer necessary to specially design the surface area of the first pre-cooling stage **25** for pre-cooling the JT cryocooler **18**, which leads to reduction in the cost of the JT cryocooler **18**.

[0044] In addition, in this embodiment, the stage extension component **50** is detachably attached to the first pre-cooling stage **25**. The first refrigerant pipe **44a** is not mounted on the first pre-cooling stage **25**. Therefore, the first pre-cooling stage **25** can be separated from the stage extension component **50** by releasing the stage extension component **50** from the first pre-cooling stage **25**. This is convenient, for example, as it facilitates the detachment of the pre-cooling cryocooler **20** from the JT cryocooler **18** for maintenance of the pre-cooling cryocooler **20**.

[0045] The present invention has been described above based on the examples. It will be understood by those skilled in the art that the present invention is not limited to the above-described embodiment, various design changes are possible, various modification examples are possible, and such modification examples are also within the scope of the present invention. Various characteristics described in relation to one embodiment are also applicable to other embodiments. A new embodiment generated through combination also has the effects of each of the combined embodiments.

[0046] FIG. **2** is a diagram schematically showing another example of the stage extension component **50** according to the embodiment. As shown in the figure, the stage extension component **50** may be attached to the second pre-cooling stage **27** of the pre-cooling cryocooler **20**. The stage extension component **50** may be thermally coupled to the second pre-cooling stage **27** and may be cooled to the second cooling temperature by the second pre-cooling stage **27**. The stage extension component **50** may be integrated with the second pre-cooling stage **27** and may function as a part of the second pre-cooling stage **27**. The stage extension component **50** may have a tubular (for example, cylindrical) shape, be fixed to the second pre-cooling stage **27** at one end, and extend in the axial direction from the second pre-cooling stage **27** to a side opposite to the second cylinder **26**.

[0047] The second refrigerant pipe **44b** may be mounted on the stage extension component **50** to enable heat exchange with the stage extension component **50**. The second refrigerant pipe **44b** connects the second heat exchanger **42b** to the third heat exchanger **42c** as described above. The second refrigerant pipe **44b** is fixed to the stage extension component **50** in a state of being wound around an outer peripheral surface of the stage extension component **50**. In this example, the second refrigerant pipe **44b** is not wound around the second pre-cooling stage **27**. The stage extension component **50** is attached and thermally coupled to the second pre-cooling stage **27**, so that the second refrigerant pipe **44b** is cooled to the second cooling temperature by the stage extension component **50** cooled by the second pre-cooling stage **27**.

[0048] In this way, the heat exchange area of the two stages of the pre-cooling cryocooler **20** can be

expanded by using the stage extension component **50**. By adjusting the shape of the stage extension component **50**, it is possible to secure an area required to sufficiently pre-cool the second refrigerant pipe **44b**. Even when the second pre-cooling stage **27** is small in size and has a small surface area, the stage extension component **50** can compensate for the lack of the heat exchange area for the second refrigerant pipe **44b**.

[0049] As in the above-described embodiment, the stage extension component **50** may be detachably attached to the second pre-cooling stage **27**. The second refrigerant pipe **44b** is not mounted on the second pre-cooling stage **27**. Therefore, the second pre-cooling stage **27** can be separated from the stage extension component **50** by releasing the stage extension component **50** from the second pre-cooling stage **27**. This is convenient, for example, as it facilitates the detachment of the pre-cooling cryocooler **20** from the JT cryocooler **18** for maintenance of the pre-cooling cryocooler **20**.

[0050] The stage extension component **50** shown in FIG. 2 may be combined with the embodiment described with reference to FIG. 1. That is, the JT cryocooler **18** may include a first stage extension component **50** attached to the first pre-cooling stage **25** and cooled by the first pre-cooling stage **25**, and a second stage extension component **50** separate from the first stage extension component **50**, attached to the second pre-cooling stage **27**, and cooled by the second pre-cooling stage **27**.

[0051] If necessary, for example, in a case where it is desired to further increase the heat exchange area, the first refrigerant pipe **44a** may be mounted on the first pre-cooling stage **25** in addition to the stage extension component **50**. Similarly, the second refrigerant pipe **44b** may be mounted on the second pre-cooling stage **27** in addition to the stage extension component **50**.

[0052] In the embodiment of FIG. 1, the stage extension component **50** extends from the first pre-cooling stage **25** toward the second pre-cooling stage **27**, but the disposition of the stage extension component **50** is not limited thereto. The stage extension component **50** may extend from the first pre-cooling stage **25** along the first cylinder **24**. In this case, the first cylinder **24** is disposed inside the stage extension component **50**. The length in the axial direction of the stage extension component **50** may be shorter than that of the first cylinder **24** so that the stage extension component **50** is not in physical contact with the vacuum chamber **12**.

[0053] Similarly, in the embodiment of FIG. 2, the stage extension component **50** may extend from the second pre-cooling stage **27** toward the first pre-cooling stage **25** along the second cylinder **26**. The second cylinder **26** is disposed inside the stage extension component **50**. The length in the axial direction of the stage extension component **50** may be shorter than that of the second cylinder **26** so that the stage extension component **50** is not in physical contact with the first pre-cooling stage **25**.

[0054] The pre-cooling cryocooler **20** is not limited to the GM cryocooler. The pre-cooling cryocooler **20** may be a cryocooler of another type such as a pulse tube cryocooler or a Stirling cryocooler.

[0055] Although the present invention has been described using specific words and phrases based on the embodiment, the embodiment merely shows one aspect of the principle and application of the present invention, and various modifications and improvements can be made within the scope of the present invention described in claims.

[0056] It should be understood that the invention is not limited to the above-described embodiment, but may be modified into various forms on the basis of the spirit of the invention. Additionally, the modifications are included in the scope of the invention.

Claims

1. A Joule-Thomson cryocooler comprising: a pre-cooling cryocooler that comprises a pre-cooling stage; a stage extension component that is attached to the pre-cooling stage and is cooled by the pre-cooling stage; and a refrigerant pipe that is mounted on the stage extension component to

enable heat exchange with the stage extension component.

2. The Joule-Thomson cryocooler according to claim 1, wherein the stage extension component is detachably attached to the pre-cooling stage.

3. The Joule-Thomson cryocooler according to claim 2, wherein the refrigerant pipe is not mounted on the pre-cooling stage.

4. The Joule-Thomson cryocooler according to claim 1, wherein the pre-cooling cryocooler comprises a cylinder extending from the pre-cooling stage, and the cylinder is disposed inside the stage extension component.

5. The Joule-Thomson cryocooler according to claim 1, wherein the pre-cooling cryocooler comprises a cylinder extending from the pre-cooling stage, and the stage extension component extends from the pre-cooling stage to a side opposite to the cylinder.

6. The Joule-Thomson cryocooler according to claim 1, wherein the pre-cooling cryocooler comprises a first pre-cooling stage and a second pre-cooling stage that is cooled to a lower temperature than the first pre-cooling stage, and the stage extension component is attached to the first pre-cooling stage and is cooled by the first pre-cooling stage.

7. The Joule-Thomson cryocooler according to claim 6, wherein the stage extension component is detachably attached to the first pre-cooling stage.

8. The Joule-Thomson cryocooler according to claim 7, wherein the refrigerant pipe is not mounted on the first pre-cooling stage.

9. The Joule-Thomson cryocooler according to claim 6, wherein the pre-cooling cryocooler comprises a cylinder that connects the second pre-cooling stage to the first pre-cooling stage, and the cylinder is disposed inside the stage extension component.

10. The Joule-Thomson cryocooler according to claim 6, wherein the stage extension component is not in physical contact with the second pre-cooling stage.
